Scoring Weights – Doctor Review

VR Medical Training Evaluation · Values currently used by the scoring engine (src/evaluation/scoring.py) · October 2026

How to use this document

The tables below list every time limit and point value the scoring engine currently uses. **All of them are placeholders** based on “Ablauf_Notfall auf Normalstation_NBA.docx” and need medical review.

- Leave **New value** empty if the current value is fine.
- Enter a different number in **New value** to change it.
- Use **Comments** for reasoning, conditions, or if a rule should be removed or worded differently.
- Rows can be added at the end of any table for rules that are missing.

> *Shaded columns are for your input. Each session starts at 100 points and the deductions below are subtracted from that.*

1. Time limits

If an action happens after this limit, a timing deduction applies (see Section 3). Times are measured from the start of the simulation.

Action	Current limit	New value	Comments
Physician / staff contact (Step 5)	10 min	2	What is STEP 1-4?
GCS / neuro check (Step 6)	5 min		Neuro Check and GCS belong to D
Full monitoring – BP, ECG, SpO2, temperature, all four (Step 7)	10 min	5	
Venous access, PVK (Step 10)	20 min	5	
Infusion started (Step 11)	22 min	6	
Blood gas analysis, BGA (Step 12)	25 min	10	Not necessary in every scenario
FAST ultrasound (Step 13)	30 min	10	Not necessary in every scenario

2. Missing actions (completeness)

Deducted when an action never happens in the session.

Missing action	Current points	New value	Comments
Airway check (A)	−10		
Breathing check (B)	−10		add as separate point: auscultation of the lungs (B)
Pulse check (C)	−10		
Capillary refill / recap time (C)	−5		
Physician contact (C)	−10		
No monitoring applied at all (C)	−15		

Missing action	Current points	New value	Comments
Each individual monitoring item never applied (BP / ECG / SpO2 / temperature)	−3 each		
Venous access (C)	−10		
Infusion (C)	−10		
Blood gas analysis (C)	−10		Not necessary in every scenario BGA does not belong to C It belongs to further diagnostics and should be performed after ABCDE (sometimes delegation to a nurse during ABCDE ist possible)
FAST ultrasound (C)	−10		Not necessary in every scenario FAST does not belong to C It belongs to further diagnostics and should be performed after ABCDE
GCS / neuro check (D)	−10		
Blood sugar – only if GCS < 15 (D)	−10		CAVE:BGA contains blood sugar
Patient undressed (E)	−10		

3. Late actions (timing)

Deducted when the action happens, but after the limit in Section 1.

Late action	Current points	New value	Comments
Physician contact late	−5		
Each monitoring item applied late	−3 each		
GCS / neuro check late	−5		
Venous access late	−5		
Infusion late	−5		
Blood gas analysis late	−5		
FAST ultrasound late	−5		

4. Wrong order (procedural violations)

The right action, but in the wrong clinical sequence. Each is penalised once per session.

Violation	Current points	New value	Comments
Adrenalin/Noradrenalin before monitoring	−10		Any medication before monitoring -> except resuscitation
Any medication before ABC assessment	−10		Exception: resuscitation

Violation	Current points	New value	Comments
Infusion prepared but not administered	−5		
FAST ultrasound before palpation	−5		
Pulse check or palpation before breathing check	−5		

5. Critical errors

Forbidden or medically unindicated actions for this scenario and location. Any critical error also triggers the safety cap (Section 6).

Critical error	Current points	New value	Comments
Wrong decision: PCI on a bleeding patient	−25		
Catecholamines (Adrenalin/Noradrenalin) on the normal ward	−20		
Unindicated anaesthesia induction (e.g. Ketamine on the ward)	−15		Add as a separate point intubation without anaesthesia induction (except resuscitation)
Shock position without prior pulse / recap check	−15		And shock position with dyspnoe oder angina pectoris
~~Oxygen given while SpO2 > 96%~~	−10		Oxygen administered to a patient with chest pain and a SpO2 > 90%

6. Pass mark and safety cap

Setting	Current value	New value	Comments
Starting score	100		
Pass threshold	70		
Safety cap – maximum score if Airway or Breathing is missed, or any critical error occurs	49 (always Fail)		

7. Open questions for the medical consultant

Question	Your answer
Should the same rules apply to all scenarios (NBA bleeding, Hypoglykämie, Gesund), or should some rules only apply to certain scenarios? (e.g. FAST ultrasound in a hypoglycaemia case)	Good question, I think we should differentiate between the scenarios Maybe we can discuss this next week
Is “patient speaking clearly” acceptable as proof of a clear airway, without an explicit airway check?	When a patient is speaking you know that he is able to breathe, but you don’t know if there is something in his mouth that should be removed → not acceptable

Question	Your answer
Should the safety cap also apply when Circulation is missed, or only Airway and Breathing?	I am not sure what you mean. We should discuss this next week
Are there important errors missing from this list? (e.g. not alerting the emergency team, leaving the patient alone)	Good question: both are important errors - intubation without capnography - not checking the airway before intubation - skipping administering oxygen to critical patient - not starting CPR, when the patient does not breathe or has a heart rate below 30/min